

AUTOSAR Engine Control Unit (ECU) High-Level Design (Version 2.0 Revision)

[SAMPLE / TEST DATA FOR REVISION COMPARISON DEMONSTRATION ONLY] | Document Version: 2.0

1. Executive Architecture Overview

This document describes the updated AUTOSAR Classic architecture for the Engine Control Unit (ECU) V2.0. Version 2.0 introduces turbocharger boost management and advanced thermal supervision components.

2. Software Component (SWC) Specification

The application layer contains components: **EngineSpeedControl_SWC**, **ThrottleControl_SWC**, and new component **TurboBoostControl_SWC**. EngineSpeedControl_SWC has been updated with new temperature sensor inputs.

SWC Name	Component Type	Provided Ports	Required Ports
EngineSpeedControl_SWC	Atomic Application SWC	PPort_EngineSpeed	RPort_ThrottlePos
ThrottleControl_SWC	Atomic Application SWC	PPort_ThrottleActuator	RPort_EngineSpeed
TurboBoostControl_SWC	Atomic Application SWC	PPort_BoostPressure	RPort_EngineSpeed

3. Basic Software (BSW) Stack Configuration

The BSW layer includes **EcuM**, **BswM**, **CanIf**, **Can**, **Com**, **Dem**, and **Dcm**. Physical CAN Driver (Can MCAL) and Diagnostic Communication Manager (Dcm) have been added in V2.0 to fulfill BSW completeness.

4. Interface & Signal Specifications

Inter-component communication relies on *SenderReceiverInterface* named **If_EngineData** and **If_BoostData**. Signal **Sig_EngineSpeed_rpm** transmits rotation velocity, **Sig_ThrottlePos_pct** transmits throttle percentage, and new signal **Sig_BoostPressure_bar** transmits manifold pressure.

5. Thermal Management Subsystem

Section 5 is added in V2.0 to define thermal protection strategies and radiator fan relay actuation.